Cloud Computing
Unit 2: Cloud Service Models

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Communication-as-a-Service (CaaS)

- Communications as a Service (CaaS) is an outsourced enterprise communications solution that can be leased from a single vendor
- Such communications can include voice over IP (VoIP or Internet telephony), instant messaging (IM), collaboration and videoconference applications using fixed and mobile devices
- The CaaS vendor is responsible for all hardware and software management and offers guaranteed Quality of Service (QoS)
- A CaaS model allows a CaaS provider's business customers to selectively deploy communications features and services throughout their company on a pay-as-you-go basis for service(s) used.

Advantages of CaaS

- From the handset found on each employee’s desk to the PC-based software client on employee laptops, to the VoIP private backbone, and all modes in between, every component in a CaaS solution is managed 24/7 by the CaaS vendor.
- Hosted and Managed Solutions
  - CaaS delivers a complete communications solution that is entirely managed by a single vendor

Fully Integrated, Enterprise-Class Unified Communications

- With CaaS, the vendor provides voice and data access and manages LAN/WAN, security, routers, email, voice mail, and data storage
- By managing the LAN/WAN, the vendor can guarantee consistent quality of service from a user's desktop across the network and back
- Advanced unified communications features that are most often a part of a standard CaaS deployment include:
  - Chat
  - Multimedia conferencing
  - Microsoft Outlook integration
  - Real-time presence
  - "Soft" phones (software-based telephones)
  - Video calling
  - Unified messaging and mobility

Fully Integrated, Enterprise-Class Unified Communications

- No Capital Expenses Needed
  - When business outsource their unified communications needs to a CaaS service provider, the provider supplies a complete solution that fits the company's exact needs. Customers pay a fee (usually billed monthly) for what they use.
  - Customers are not required to purchase equipment, so there is no capital outlay
- Flexible Capacity and Feature Set
  - When customers outsource communications services to a CaaS provider, they pay for the features they need when they need them
  - The service provider can distribute the cost services and delivery across a large customer base. This makes the use of shared feature functionality more economical for customers to implement.
  - Economies of scale allow service providers enough flexibility that they are not tied to a single vendor investment

Fully Integrated, Enterprise-Class Unified Communications

- No Risk of Obsolescence
  - The average life cycles key communications equipment and systems range anywhere from five to 10 years.
  - With the constant introduction of newer models for all sorts of technology (PCs, cell phones, video software and hardware, etc.), these types of products now face much shorter life cycles, sometimes as short as a single year
  - CaaS vendors must absorb this burden for the user by continuously upgrading the equipment in their offerings to meet changing demands in the marketplace.
- No Facilities and Engineering Costs Incurred
  - CaaS providers host all of the equipment needed to provide their services to their customers, virtually eliminating the need for customers to maintain data center space and facilities
  - There is no extra expense for the constant power consumption that such a facility would demand
  - Customers receive the benefit of multiple carrier-grade data centers with full redundancy—and it’s all included in the monthly payment

Fully Integrated, Enterprise-Class Unified Communications

- Guaranteed Business Continuity
  - Distributing risk by using geographically dispersed data centers has become the norm today
  - It mitigates risk and allows companies in a location hit by a catastrophic event to recover as soon as possible
  - This process is implemented by CaaS providers because most companies don’t even contemplate voice continuity if catastrophe strikes

Infrastructure-as-a-Service (IaaS)

- Infrastructure-as-a-Service (IaaS) is the delivery of computer infrastructure (typically a platform virtualization environment) as a service
- IaaS leverages significant technology, services, and data center investments to deliver IT as a service to customers
- Unlike traditional outsourcing, which requires extensive due diligence, negotiations ad infinitum, and complex, lengthy contract vehicles, IaaS is centered around a model of service delivery that provisions a predefined, standardized infrastructure specifically optimized for the customer’s applications.

Infrastructure-as-a-Service (IaaS)

- Infrastructure as a Service (IaaS) includes the capability to provision processing, storage, networks, and other fundamental computing resources; the consumer is able to deploy and run arbitrary software, which can include operating systems and applications.
- The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems; storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).
- IaaS clouds often offer additional resources such as; images in a virtual machine image library, raw (block) and file-based storage, firewalls, IP addresses, virtual local area networks (VLANs) and software bundles

Infrastructure-as-a-Service (IaaS)

- IaaS providers manage the transition and hosting of selected applications on their infrastructure
- Provider-owned implementations typically include the following layered components:
  - Computer hardware (typically set up as a grid for massive horizontal scalability)
  - Computer network (including routers, firewalls, load balancing, etc.)
  - Internet connectivity
  - Platform virtualization environment for running client-specified virtual machines
  - Service-level agreements
  - Utility computing billing

Infrastructure-as-a-Service (IaaS)

- The chief benefits of using this type of outsourced service include:
  - Ready access to a preconfigured environment that is generally ITIL-based (The Information Technology Infrastructure Library [ITIL] is a customized framework of best practices designed to promote quality computing services in the IT sector.)
  - Use of the latest technology for infrastructure equipment
  - Secured, “sand-boxed” (protected and insulated) computing platforms that are usually security monitored for breaches
  - Reduced risk by having off-site resources maintained by third parties
  - Ability to manage service-demand peaks and valleys
  - Lower costs that allow expensing service costs instead of making capital investments
  - Reduced time, cost, and complexity in adding new features or capabilities

Infrastructure-as-a-Service (IaaS)

- Some of the most popular IaaS solutions are:
  - Compute as a service: One of the most ubiquitous IaaS offerings today, compute as a service provides compute capacity that includes servers, operating system access, firewalls, routers and load balancing on demand.
  - Web hosting: Moving a website to an IaaS based model ensures that the website won't get bogged down during peak traffic times
  - Storage as Service: Storage as a Service is a business model in which a large company rents space in their storage infrastructure to a smaller company or individual.
  - Disaster recovery and backup as a service: The idea behind moving disaster recovery to the cloud is to ensure that organizations have uninterrupted access to data and applications, regardless of emergencies, such as power outages, natural disasters or system failures.

Infrastructure-as-a-Service (IaaS)

- Some of the most popular IaaS solutions are:
  - Desktops as a service: DaaS is a way to make sure that there are always enough desktop environments available to new workers, with enough storage and all the right applications. And because the desktops can be accessed via the Internet, users can log in and access their familiar workspaces from any location.
  - Servers as a service: Accessing servers in the cloud means that no matter what the project, or even if it’s the busy season, there will always be enough compute power to go around.
  - Networking as a service: The idea is to offer networking resources on demand in order to support virtual networks — resources such as firewalls, load balancing and WAN acceleration services.

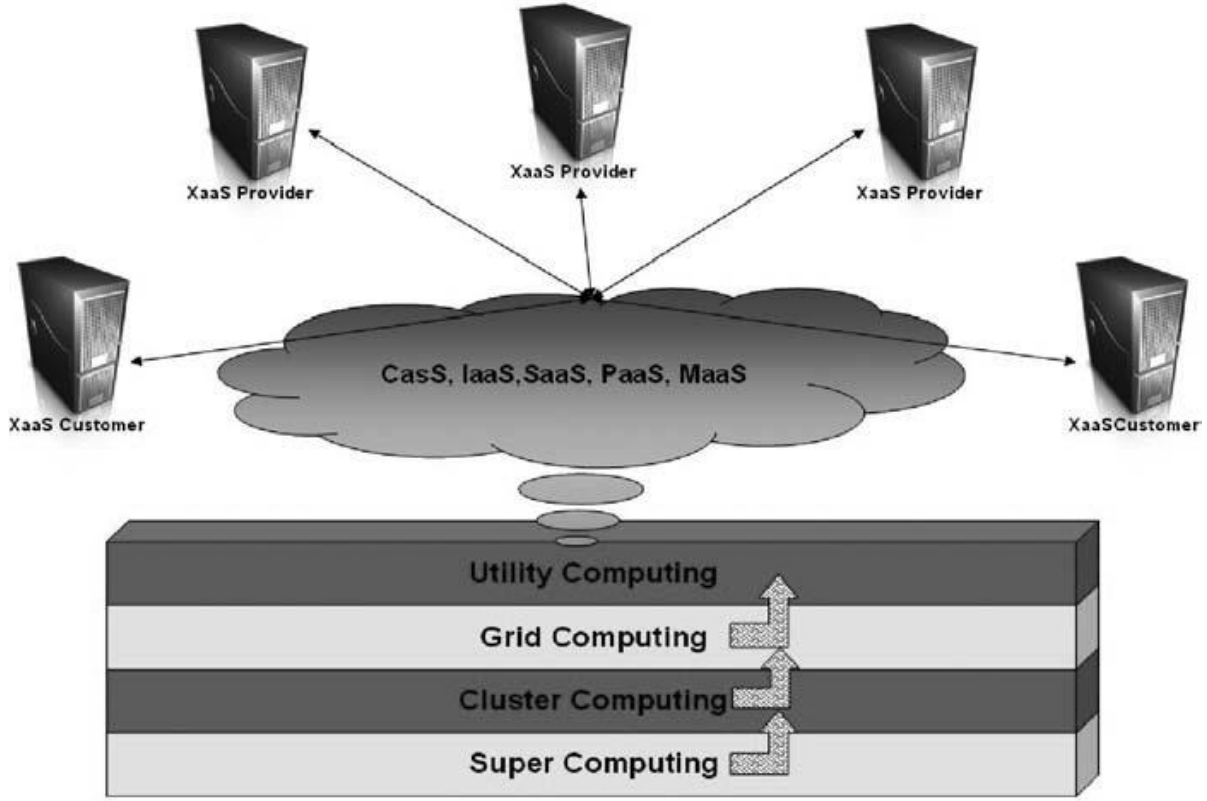
Modern On-Demand Computing

- On-demand computing is an increasingly popular enterprise model in which computing resources are made available to the user as needed
- Computing resources that are maintained on a user's site are becoming fewer and fewer, while those made available by a service provider are on the rise
- The on-demand model evolved to overcome the challenge of being able to meet fluctuating resource demands efficiently
- Because demand for computing resources can vary drastically from one time to another, maintaining sufficient resources to meet peak requirements can be costly

Modern On-Demand Computing

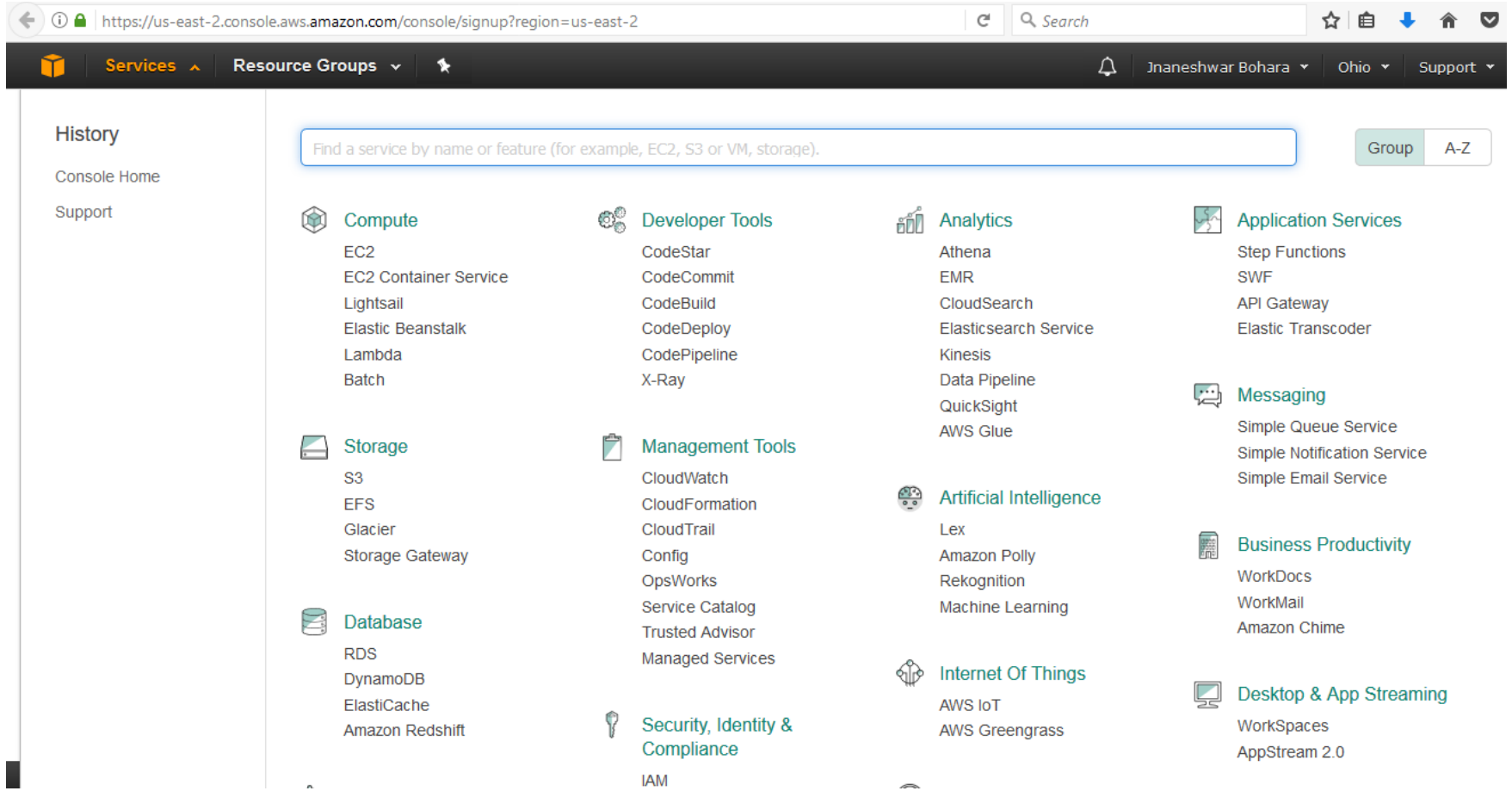
- Concepts such as clustered computing, grid computing, utility computing, etc., may all seem very similar to the concept of on-demand computing
- But they can be better understood if one thinks of them as building blocks that evolved over time and with techno-evolution to achieve the modern cloud computing model we think of and use today
- One example that we can examine is Amazon's Elastic Compute Cloud (Amazon EC2). This is a web service that provides resizable computing capacity in the cloud.

Modern On-Demand Computing



Building blocks to the cloud

Amazon’s Elastic Cloud



Amazon’s Elastic Cloud

Create a New Instance Cancel

Select an option below:

Classic Wizard
Launch an On-Demand or Spot instance using the classic wizard with fine-grained control over how it is launched.

Quick Launch Wizard
Launch an On-Demand instance using an editable, default configuration so that you can get started in the cloud as quickly as possible.

AWS Marketplace
AWS Marketplace is an online store where you can find and buy software that runs on AWS. Launch with 1-Click and pay by the hour.

Submit Feedback Getting Started Guide

Name Your Instance: GSG Tutorial Pick a meaningful name, e.g. Web Server

Choose a Key Pair:
Public/private key pairs allow you to securely connect to your instance after it launches.
Select Existing | Create New | None

Name: GSG_Keypair Download
Please note that you need to download the key pair before you can continue.

Choose a Launch Configuration:

More Amazon Machine Images NEW!
Search through public and AWS Marketplace AMIs or choose from your own custom AMIs.

Amazon Linux AMI 2012.03 64 bit 32 bit
The Amazon Linux AMI 2012.03 is an EBS-backed, PV-GRUB image. It includes Linux 3.2, AWS tools, and repository access to multiple versions of MySQL, PostgreSQL, Python, Ruby, and Tomcat. Free tier eligible

Red Hat Enterprise Linux 6.3 64 bit 32 bit
Red Hat Enterprise Linux version 6.3, EBS-boot.

SUSE Linux Enterprise Server 11 64 bit 32 bit
SUSE Linux Enterprise Server 11 Service Pack 2 basic install, EBS boot with Amazon EC2 AMI Tools preinstalled; Apache 2.2, MySQL 5.0, PHP 5.3, and Ruby 1.8.7

Ubuntu Server 12.04 LTS 64 bit 32 bit
Ubuntu Server 12.04 LTS with support available from Canonical (http://www.ubuntu.com/cloud/services). Free tier eligible

Note: You can customize your settings in the next step. Continue

What is Amazon Web Services ?

- **Amazon Web Services (AWS)** is a collection of **remote computing services (web services)** that together make up a **cloud computing platform**, offered over the Internet by Amazon.com.
- Website: http://aws.amazon.com
- AWS is located in **9 geographical 'Regions**'. Each Region is **wholly contained within a single country** and all of its data and services **stay** within the designated Region.
- Each Region has **multiple 'Availability Zones**', which are **distinct data centers** providing AWS services. Availability Zones are **isolated from each other** to prevent outages from spreading between Zones. However, Several services **operate across** Availability Zones (e.g. S3, DynamoDB).

Amazon’s Elastic Cloud

- Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud
- It is designed to make web-scale cloud computing easier for developers
- Amazon EC2’s simple web service interface allows you to obtain and configure capacity with minimal effort
- It provides you with complete control of your computing resources and lets you run on Amazon’s proven computing environment
- Amazon EC2 reduces the time required to obtain and boot new server instances to minutes, allowing you to quickly scale capacity, both up and down, as your computing requirements change

Getting Started with Amazon EC2

- Step 1: Sign up for Amazon EC2
- Step 2: Create a key pair
- Step 3: Launch an Amazon EC2 instance
- Step 4: Connect to the instance
- Step 5: Customize the instance
- Step 6: Terminate instance and delete the volume created

Terminology

- Instance = One running virtual machine.
- Instance Type = hardware configuration: cores, memory, disk.
- Instance Store Volume = Temporary disk associated with instance.
- Image (AMI) = Stored bits which can be turned into instances.
- Key Pair = Credentials used to access VM from command line.
- Region = Geographic location, price, laws, network locality.
- Availability Zone = Subdivision of region the is fault-independent.

Amazon Elastic Compute Cloud (EC2)

- A web service that provides **resizable compute capacity** in the cloud.
- EC2 allows **creating Virtual Machines (VM) on-demand**. Pre-configured **templated Amazon Machine Image (AMI)** can be used get running immediately. Creating and sharing your own AMI is also possible via the **AWS Marketplace.**
- **Auto Scaling** allows **automatically scale of the capacity up** seamlessly during **demand spikes** to maintain performance, and **scales down** during **demand lulls** to minimize costs.
- **Elastic Load Balancing** automatically distributes incoming application traffic across multiple Amazon EC2 instances.
- Provide tools to build **failure resilient applications** by launching application instances in **separate Availability Zones**.
- Pay only for resources actually consume, **instance-hours.**
- **VM Import/Export** enables you to easily import virtual machine images from your existing environment to Amazon EC2 instances and export them back at any time.

Amazon Elastic Compute Cloud (EC2)

- Amazon Machine Images (**AMIs**) are the basic building blocks of Amazon EC2
- An AMI is a template that contains a software configuration (operating system, application server and applications) that can run on Amazon's computing environment
- AMIs can be used to launch an ***instance***, which is a copy of the AMI running as a virtual server in the cloud.

Region and Zones

- AWS is located in **9 geographical 'Regions**‘. Each Region is **wholly contained within a single country** and all of its data and services **stay** within the designated Region.
- Each Region has **multiple 'Availability Zones'**, which are **distinct data centers** providing AWS services. Availability Zones are **isolated from each other** to prevent outages from spreading between Zones. However, Several services **operate across** Availability Zones (e.g. S3, DynamoDB).

EC2 Payment methods

- **On-Demand Instances** let you **pay for compute capacity by the hour** with **no long-term commitments**.
- **Reserved Instances** give you the option to **make a low, one-time payment for each instance** you want to reserve and in turn **receive a significant discount on the hourly charge** for that instance.
- **Spot Instances** allow customers to **bid on unused Amazon EC2 capacity** and **run those instances for as long as their bid exceeds the current Spot Price**.

Storage

- Amazon EC2 provides three type of storage option
  - Amazon EBS
  - Amazon S3
  - Instance Storage

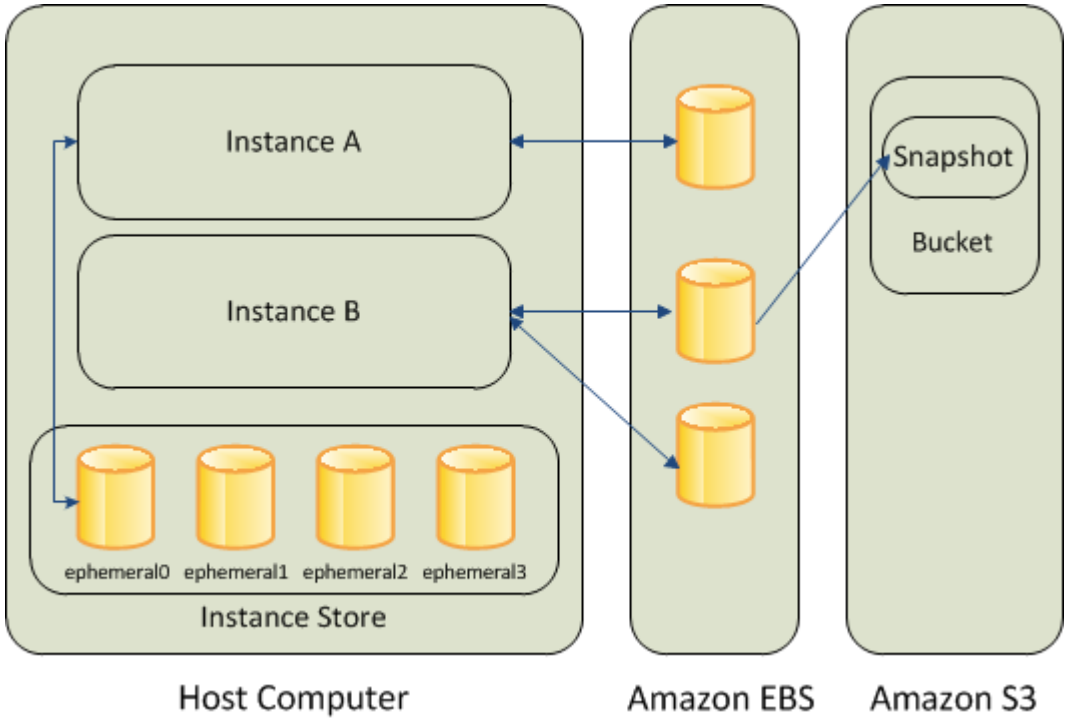


Diagram Source: http://docs.aws.amazon.com

Amazon Elastic Block Store (EBS)

- Provides **block level storage** volumes (**1 GB to 1 TB**) for **use with Amazon EC2 instances**.
  - **Multiple volumes** can be mounted to the **same instance**.
  - EBS volumes are **network-attached**, and **persist independently** from the life of an instance.
  - Storage volumes behave like **raw, unformatted block devices**, allowing users to **create a file system** on top of Amazon EBS volumes, or use them in any other way you would use a block device (**like a hard drive**).
- EBS volumes are **placed in a specific Availability Zone**, and can then be **attached to instances also in that same Availability Zone**.
- Each storage volume is **automatically replicated within the same Availability Zone**.
- EBS provides the ability to **create point-in-time snapshots of volumes**, which are **persisted to Amazon S3**.
  - These snapshots can be **used as the starting point for new Amazon EBS volumes**, and protect data for long-term durability.
  - The **same snapshot can be used to instantiate as many volumes** as you wish.
  - These snapshots **can be copied across AWS regions**.

Amazon Simple Storage Service (S3)

- Amazon S3 provides a simple web services interface that can be **used to store and retrieve any amount of data, at any time, from anywhere on the web**.
- Write, read, and delete objects containing from **1 byte to 5 terabytes of data each**. The **number of objects** you can store is **unlimited**.
- Each object is stored in a **bucket** and retrieved via a unique, developer-assigned **key**.
  - A bucket can be stored in **one of several Regions**.
  - You can **choose a Region** to **optimize for latency, minimize costs, or address regulatory requirements**.
  - Objects stored in a Region **never leave the Region** unless you transfer them out.
- **Authentication mechanisms** are provided to ensure that data is kept secure from unauthorized access.
  - Objects can be made **private or public, and rights can be granted to specific users**.
- S3 charges based on **per GB-month** AND **per I/O requests** AND **per data modification requests.**

Networking and Security

- Instances can be launched on one of the two platforms
  - EC2-Classic
  - EC2-VPC
- Instance IP address is dynamic.
  - new IP address is assigned every time instance is launched
- Static IP address – Elastic IP address
  - Remap the Elastic IP to new instance to mask failure
  - Separate pool for EC2-Classic and VPC
- Security Groups to access control to instance

Monitoring, Auto Scaling, and Load Balancing

- Monitor statistics of instances and EBS
  - CloudWatch
- Automatically scales amazon EC2 capacity up and down based on rules
  - Add and remove compute resource based on demand
  - Suitable for businesses experiencing variability in usage
- Distribute incoming traffic across multiple instances
  - Elastic Load Balancing

Amazon EC2 Service Characteristics

- Dynamic Scalability
  - Amazon EC2 enables users to increase or decrease capacity in a few minutes
  - Users can invoke a single instance, hundreds of instances, or even thousands of instances simultaneously
  - Because this is all controlled with web service APIs, an application can automatically scale itself up or down depending on its needs
- Full Control of Instances
  - Users have complete control of their instances
  - They have root access to each instance and can interact with them as one would with any machine
  - Instances can be rebooted remotely using web service APIs

Amazon EC2 Service Characteristics

- Configuration Flexibility
  - Configuration settings can vary widely among users
  - They have the choice of multiple instance types, operating systems, and software packages
  - Amazon EC2 allows them to select a configuration of memory, CPU, and instance storage that is optimal for their choice of operating system and application
- Integration with Other Amazon Web Services
  - Amazon EC2 works in conjunction with a variety of other Amazon web services
  - Amazon Simple Storage Service (Amazon S3), Amazon SimpleDB, Amazon Simple Queue Service (Amazon SQS), and Amazon CloudFront are all integrated to provide a complete solution for computing, query processing, and storage across a wide range of applications

Amazon EC2 Service Characteristics

- Reliable and Resilient Performance
  - Amazon Elastic Block Store (EBS) is yet another Amazon EC2 feature that provides users powerful features to build failure-resilient applications
  - Amazon EBS offers persistent storage for Amazon EC2 instances
  - Amazon EBS volumes provide “off-instance” storage that persists independently from the life of any instance
- Support for Use in Geographically Disparate Locations
  - Amazon EC2 provides users with the ability to place one or more instances in multiple locations
  - By launching instances in any one or more of the separate Availability Zones, one can insulate their applications from a single point of failure

Elastic IP Addressing

- Elastic IP (EIP) addresses are static IP addresses designed for dynamic cloud computing
- An Elastic IP address is associated with your account and not with a particular instance, and you control that address until you choose explicitly to release it
- Unlike traditional static IP addresses, however, EIP addresses allow you to mask instance or Availability Zone failures by programmatically remapping your public IP addresses to any instance in your account
- Rather than waiting on a technician to reconfigure or replace your host, or waiting for DNS to propagate to all of your customers, Amazon EC2 enables you to work around problems that occur with client instances or client software by quickly remapping their EIP address to another running instance.
- A significant feature of Elastic IP addressing is that each IP address can be reassigned to a different instance when needed

How the Elastic IPs work with Amazon EC2 services

- First of all, Amazon allows users to allocate up to five Elastic IP addresses per account (which is the default)
- Each EIP can be assigned to a single instance
- When this reassignment occurs, it replaces the normal dynamic IP address used by that instance
- By default, each instance starts with a dynamic IP address that is allocated upon startup
- Since each instance can have only one external IP address, the instance starts out using the default dynamic IP address
- If the EIP in use is assigned to a different instance, a new dynamic IP address is allocated to the vacated address of that instance

How the Elastic IPs work with Amazon EC2 services

- Assigning or reassigning an IP to an instance requires only a few minutes
- The limitation of designating a single IP at a time is due to the way Network Address Translation (NAT) works
- Each instance is mapped to an internal IP address and is also assigned an external (public) address
- The public address is mapped to the internal address using Network Address Translation tables (hence, NAT)

Monitoring-as-a-Service (MaaS)

- Monitoring-as-a-Service (MaaS) is the outsourced provisioning of security, primarily on business platforms that leverage the Internet to conduct business
- Security monitoring involves protecting an enterprise or government client from cyber threats
- A security team plays a crucial role in securing and maintaining the confidentiality, integrity, and availability of IT assets
- However, time and resource constraints limit security operations and their effectiveness for most companies
- This requires constant vigilance over the security infrastructure and critical information assets

Monitoring-as-a-Service (MaaS)

- Many industry regulations require organizations to monitor their security environment, server logs, and other information assets to ensure the integrity of these systems
- Conducting effective security monitoring can be a daunting task because it requires advanced technology, skilled security experts, and scalable processes—none of which come cheap
- MaaS security monitoring services offer real-time, 24/7 monitoring and nearly immediate incident response across a security infrastructure—they help to protect critical information assets of their customers

Protection Against Internal and External Threats

- Typical services provided by many MaaS vendors
  - Early Detection
  - Platform, Control, and Services Monitoring
  - Intelligent Log Centralization and Analysis
  - Vulnerabilities Detection and Management
  - Continuous System Patching/Upgrade and Fortification
  - Intervention, Forensics, and Help Desk Services

Protection Against Internal and External Threats

- Early Detection

- An early detection service detects and reports new security vulnerabilities shortly after they appear
- Generally, the threats are correlated with third party sources, and an alert or report is issued to customers
- This report is usually sent by email to the person designated by the company
- Security vulnerability reports, aside from containing a detailed description of the vulnerability and the platforms affected, also include information on the impact the exploitation of this vulnerability would have on the systems or applications previously selected by the company receiving the report
- Most often, the report also indicates specific actions to be taken to minimize the effect of the vulnerability, if that is known.

Protection Against Internal and External Threats

- Platform, Control, and Services Monitoring
  - Platform, control, and services monitoring is often implemented as a dashboard interface and makes it possible to know the operational status of the platform being monitored at any time
  - Each operational element that is monitored usually provides an operational status indicator, always taking into account the critical impact of each element
  - This service aids in determining which elements may be operating at or near capacity or beyond the limits of established parameters
  - By detecting and identifying such problems, preventive measures can be taken to prevent loss of service

Protection Against Internal and External Threats

- Intelligent Log Centralization and Analysis
  - Intelligent log centralization and analysis is a monitoring solution based mainly on the correlation and matching of log entries
  - Such analysis helps to establish a baseline of operational performance and provides an index of security threat
  - Alarms can be raised in the event an incident moves the established baseline parameters beyond a stipulated threshold
  - These types of sophisticated tools are used by a team of security experts who are responsible for incident response once such a threshold has been crossed and the threat has generated an alarm or warning picked up by security analysts monitoring the systems

Protection Against Internal and External Threats

- Vulnerabilities Detection and Management
  - Vulnerabilities detection and management enables automated verification and management of the security level of information systems
  - The service periodically performs a series of automated tests for the purpose of identifying system weaknesses that may be exposed over the Internet, including the possibility of unauthorized access to administrative services, the existence of services that have not been updated, the detection of vulnerabilities such as phishing, etc
  - The service performs periodic follow-up of tasks performed by security professionals managing information systems security and provides reports that can be used to implement a plan for continuous improvement of the system's security level.

Protection Against Internal and External Threats

- Continuous System Patching/Upgrade and Fortification
  - Security posture is enhanced with continuous system patching and upgrading of systems and application software
  - New patches, updates, and service packs for the equipment's operating system are necessary to maintain adequate security levels and support new versions of installed products

Protection Against Internal and External Threats

- Intervention, Forensics, and Help Desk Services
  - Quick intervention when a threat is detected is crucial to mitigating the effects of a threat
  - This requires security engineers with ample knowledge in the various technologies and with the ability to support applications as well as infrastructures on a 24/7 basis
  - MaaS platforms routinely provide this service to their customers
  - When a detected threat is analyzed, it often requires forensic analysis to determine what it is, how much effort it will take to fix the problem, and what effects are likely to be seen
  - Help desk services provide assistance on questions or issues about the operation of running systems

Platform-as-a-Service (PaaS)

- Cloud computing has evolved to include platforms for building and running custom web-based applications, a concept known as Platform-as-a- Service
- The PaaS model makes all of the facilities required to support the complete life cycle of building and delivering web applications and services entirely available from the Internet, all with no software downloads or installation for developers, IT managers, or end users
- PaaS developers are concerned only with web based development and generally do not care what operating system is used
- Anyone with an Internet connection can build powerful applications and easily deploy them to users globally

Platform-as-a-Service (PaaS)

- PaaS provides the capability for consumers to have applications deployed without the burden and cost of buying and managing the hardware and software
- PaaS facilitates immediate business requirements such as application design, development and testing at a fraction of the normal cost

The Traditional On-Premises Model

- The traditional approach of building and running on-premises applications has always been complex, expensive, and risky
- Each application was designed to meet specific business requirements.
- Each solution required a specific set of hardware, an operating system, a database, often a middleware package, email and web servers, etc
- Once the hardware and software environment was created, a team of developers had to navigate complex programming development platforms to build their applications

The Traditional On-Premises Model

- Additionally, a team of network, database, and system management experts was needed to keep everything up and running
- Inevitably, a business requirement would force the developers to make a change to the application
- The changed application then required new test cycles before being distributed
- Large companies often needed specialized facilities to house their data centers

The New Cloud Model : PaaS

- PaaS offers a faster, more cost-effective model for application development and delivery
- PaaS provides all the infrastructure needed to run applications over the Internet
- PaaS offerings include workflow facilities for application design, application development, testing, deployment, and hosting, as well as application services such as virtual offices, team collaboration, database integration, security, scalability, storage, persistence, state management, dashboard instrumentation, etc.

Key Characteristics of PaaS

- Chief characteristics of PaaS include services to develop, test, deploy, host, and manage applications to support the application development life cycle
- PaaS providers often include services for concurrency management, scalability, fail-over and security
- Another characteristic is the integration with web services and databases
- Support for Simple Object Access Protocol (SOAP) and other interfaces allows PaaS offerings to create combinations of web services (called mashups) as well as having the ability to access databases and reuse services maintained inside private networks

Key Characteristics of PaaS

- Ability to form and share code with ad-hoc, predefined, or distributed teams greatly enhance the productivity of PaaS offerings
- Integrated PaaS offerings **provide an opportunity for developers to have much greater insight into the inner workings of their applications and the behavior of their users by implementing dashboard- like tools to view the inner workings based on measurements such as performance, number of concurrent accesses**, etc. Some PaaS offerings leverage this instrumentation to enable pay-per-use billing models

Software as a service (Saas)

Traditional Software

Build Your Own

On-Demand Utility

Plug In, Subscribe

Pay-per-Use

Comparison of business model

Traditional packaged software	Software as a serivce
Designed for customers to install, manage and maintain.	Designed from the outset up for delivery as Internet-based services
Architect solutions to be run by an individual company in a dedicated instantiation of the software	Designed to run thousands of different customers on a single code

Traditional packaged Software

- Infrequent, major upgrades every 18-24 months, sold individually to each installed base customer.
- Version control
- Upgrade fee
- Streamlined, repeatable functionality via Web services, open APIs and standard connectors

Software as a serivce

- Frequent, "digestible" upgrades every 3-6 months to minimize customer disruption and enhance satisfaction.
- Fixing a problem for one customer fixes it for everyone
- May use open APIs and Web services to facilitate integration, but each customer must typically pay for one-off integration work.

Software-as-a-Service (SaaS)

- The traditional model of software distribution, in which software is purchased for and installed on personal computers, is sometimes referred to as Software-as-a-Product
- Software-as-a-Service is a software distribution model in which applications are hosted by a vendor or service provider and made available to customers over a network, typically the Internet
- SaaS is becoming an increasingly prevalent delivery model as underlying technologies that support web services and service-oriented architecture (SOA) mature and new developmental approaches become popular
- SaaS is also often associated with a pay-as-you-go subscription licensing model

Software-as-a-Service (SaaS)

- International Data Corportation identifies two slightly different delivery models for SaaS
- The hosted application management model is similar to an Application Service Provider (ASP) model. Here, an ASP hosts commercially available software for customers and delivers it over the Internet
- The other model is a software on demand model where the provider gives customers network-based access to a single copy of an application created specifically for SaaS distribution

Software-as-a-Service (SaaS)

- SaaS is most often implemented to provide business software functionality to enterprise customers at a low cost while allowing those customers to obtain the same benefits of commercially licensed, internally operated software without the associated complexity of installation, management, support, licensing, and high initial cost
- Many types of software are well suited to the SaaS model (e.g., accounting, customer relationship management, email software, human resources, IT security, IT service management, video conferencing, web analytics, web content management).
- The distinction between SaaS and earlier applications delivered over the Internet is that SaaS solutions were developed specifically to work within a web browser
- The architecture of SaaS-based applications is specifically designed to support many concurrent users (multitenancy) at once

SaaS Implementation Issues

- According to Microsoft, SaaS architectures can be classified into one of four maturity levels whose key attributes are ease of configuration, multitenant efficiency, and scalability
- Each level is distinguished from the previous one by the addition of one of these three attributes

SaaS Implementation Issues

- SaaS Architectural Maturity Level 1—Ad-Hoc/Custom
  - The first level of maturity is actually no maturity at all
  - Each customer has a unique, customized version of the hosted application
  - The application runs its own instance on the host's servers
- SaaS Architectural Maturity Level 2—Configurability
  - The second level of SaaS maturity provides greater program flexibility through configuration metadata
  - At this level, many customers can use separate instances of the same application
  - This allows a vendor to meet the varying needs of each customer by using detailed configuration options
  - It also allows the vendor to ease the maintenance burden by being able to update a common code base

SaaS Implementation Issues

- SaaS Architectural Maturity Level 3—Multitenant Efficiency
  - The third maturity level adds multitenancy to the second level
  - This results in a single program instance that has the capability to serve all of the vendor's customers
  - This approach enables more efficient use of server resources without any apparent difference to the end user, but ultimately this level is limited in its ability to scale massively
- SaaS Architectural Maturity Level 4—Scalable
  - At the fourth SaaS maturity level, scalability is added by using a multitiered architecture
  - This architecture is capable of supporting a load-balanced farm of identical application instances running on a variable number of servers, sometimes in the hundreds or even thousands
  - System capacity can be dynamically increased or decreased to match load demand by adding or removing servers, with no need for further alteration of application software architecture

Key characteristics of SaaS

- Network-based management and access to commercially available software from central locations rather than at each customer's site, enabling customers to access applications remotely via the Internet
- Application delivery from a one-to-many model (single-instance, multitenant architecture), as opposed to a traditional one-to-one model
- Centralized enhancement and patch updating that obviates any need for downloading and installing by a user. SaaS is often used in conjunction with a larger network of communications and collaboration software, sometimes as a plug-in to a PaaS architecture

Benefits of SaaS Model

- Streamlined administration
- Automated update and patch management services
- Data compatibility across the enterprise (all users have the same version of software)
- Facilitated, enterprise-wide collaboration
- Global accessibility

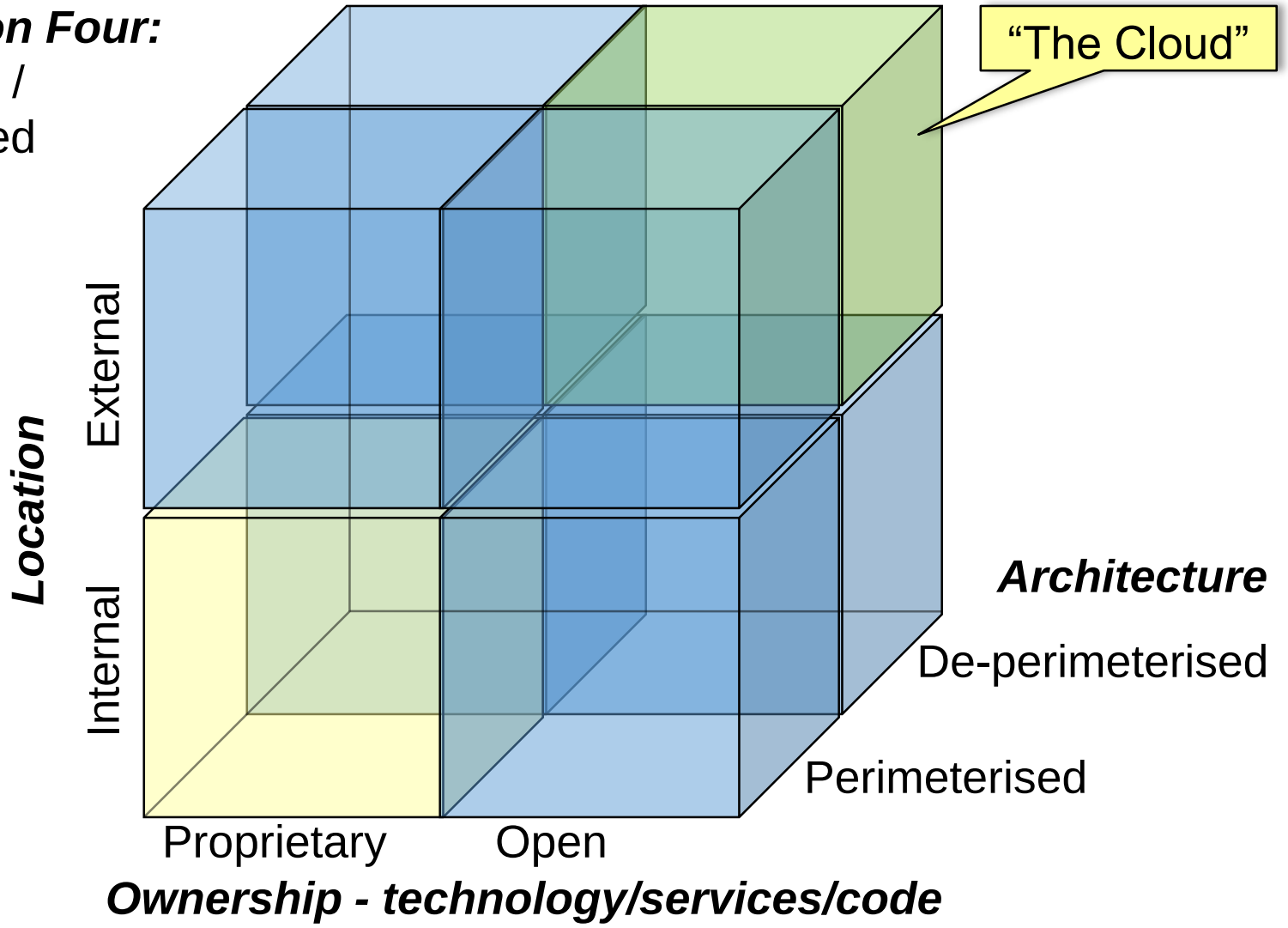
SaaS - Pros

- Stay focused on business processes
- Change software to an Operating Expense instead of a Capital Purchase, making better accounting and budgeting sense.
- Create a consistent application environment for all users
- No concerns for cross platform support
- Easy Access
- Reduced piracy of your software
- Lower Cost
  - For an affordable monthly subscription
  - Implementation fees are significantly lower
- Continuous Technology Enhancements

SaaS - Cons

- Initial time needed for licensing and agreements
  - *Trust*, or the lack thereof, is the number one factor blocking the adoption of software as a service (SaaS).
  - Centralized control
  - Possible erosion of customer privacy
- Absence of disconnected use

Jericho Cloud Cube Model

Jericho Forum Cloud Cube Model

Dimension Four:
Insourced /
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